CHAPTER 5

Minerals

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What is a mineral

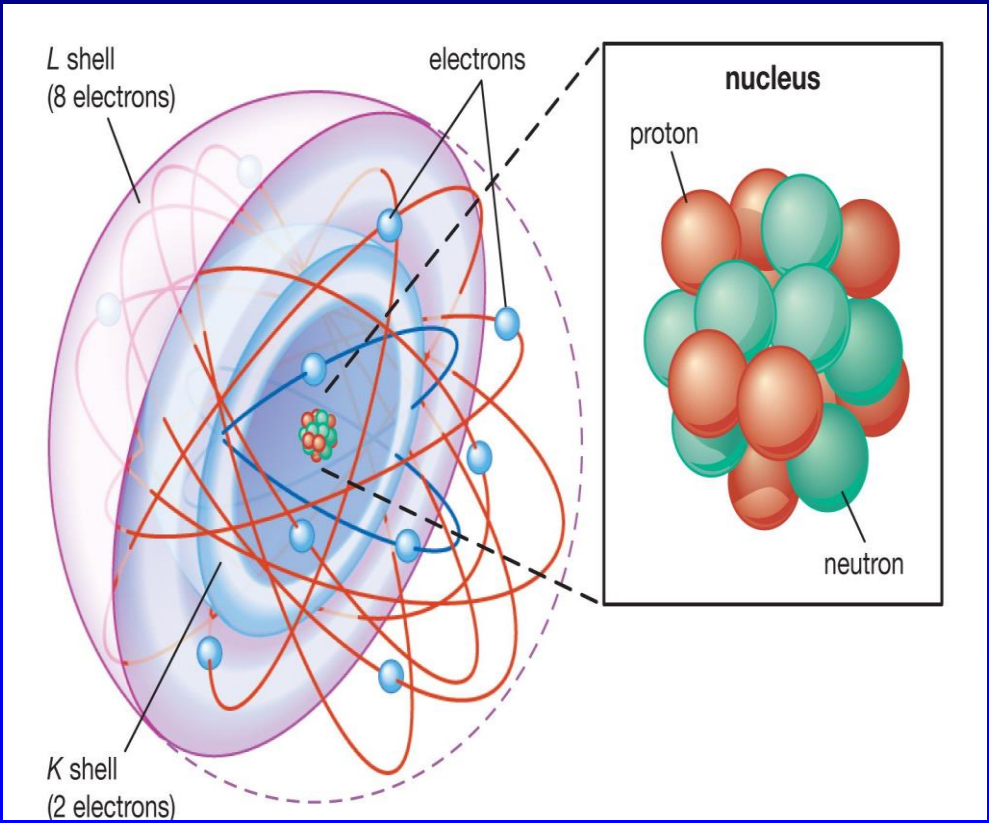
- a naturally occurring
- inorganic solid
- Fixed chemical composition
- orderly internal structure

Elements, atoms, and the chemical composition of minerals

-Every solid, liquid and gas, is composed of neutral or ionized atoms. Atoms are very small; typical sizes are around 100 pm.

***Atoms** represent the basic building blocks of the elements and the atoms of different elements combine to form minerals.*

Atom has some protons and neutrons in the nucleus and some electrons zipping around in orbitals.



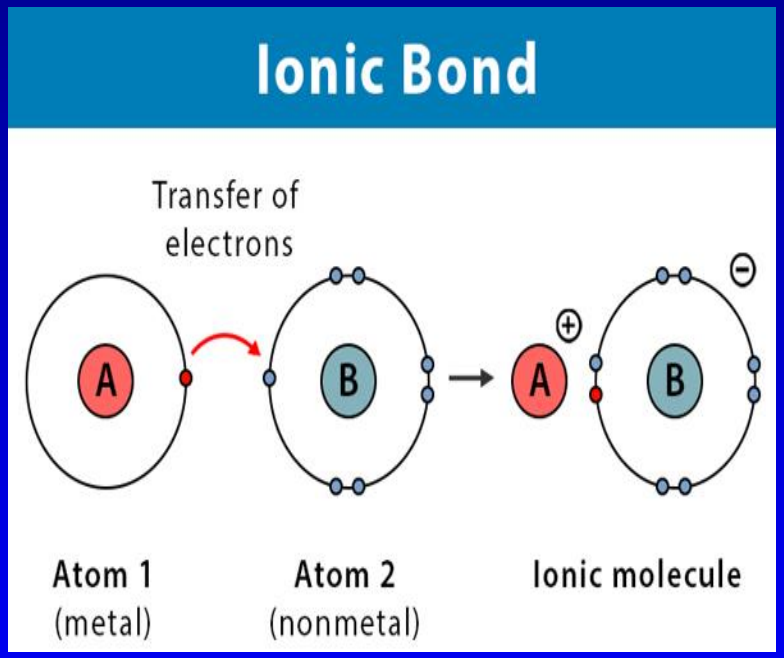
When an atom (or group of atoms) gains or losses one or more electrons, it forms an ***ion****.*

Ions of individual elements combine together to form mineral compounds by chemical bonding.

Chemical bonding.

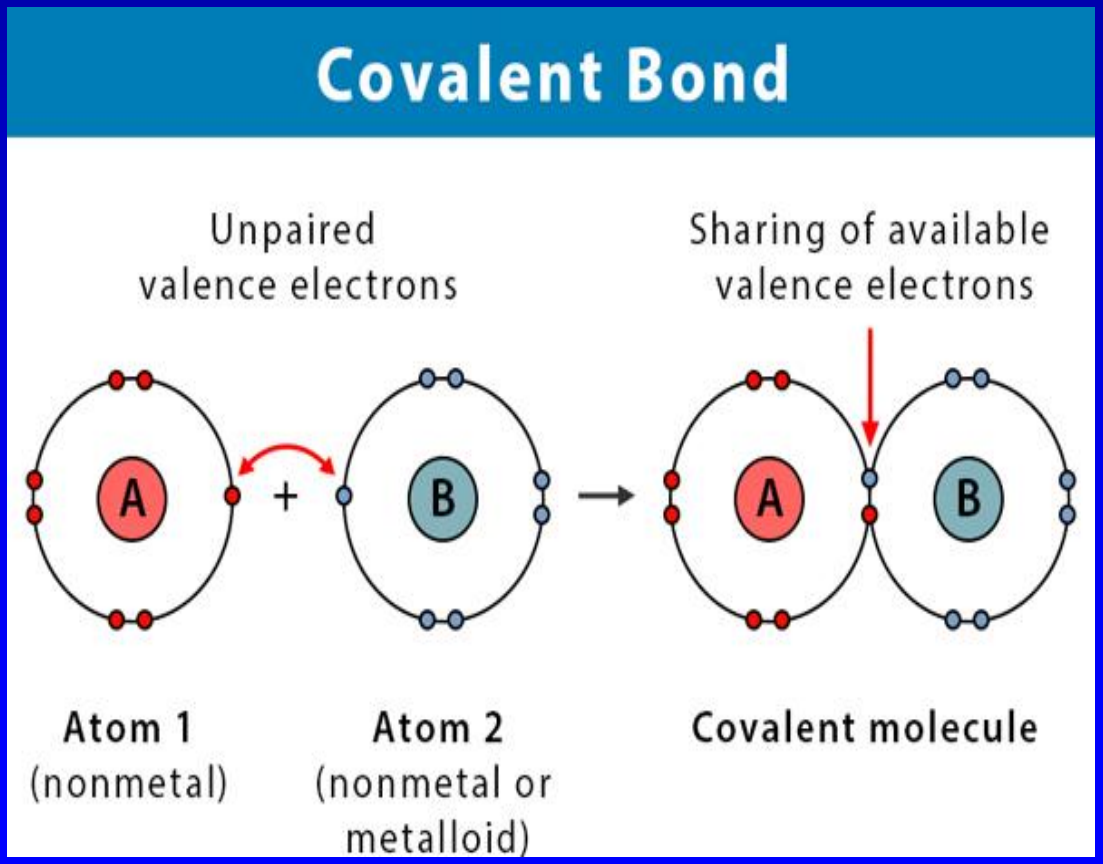
1-Ionic Bond:رابطة ايونية

Ionic bond is typically described to students as being the outcome of the transfer of electron (s) between two dissimilar atoms.



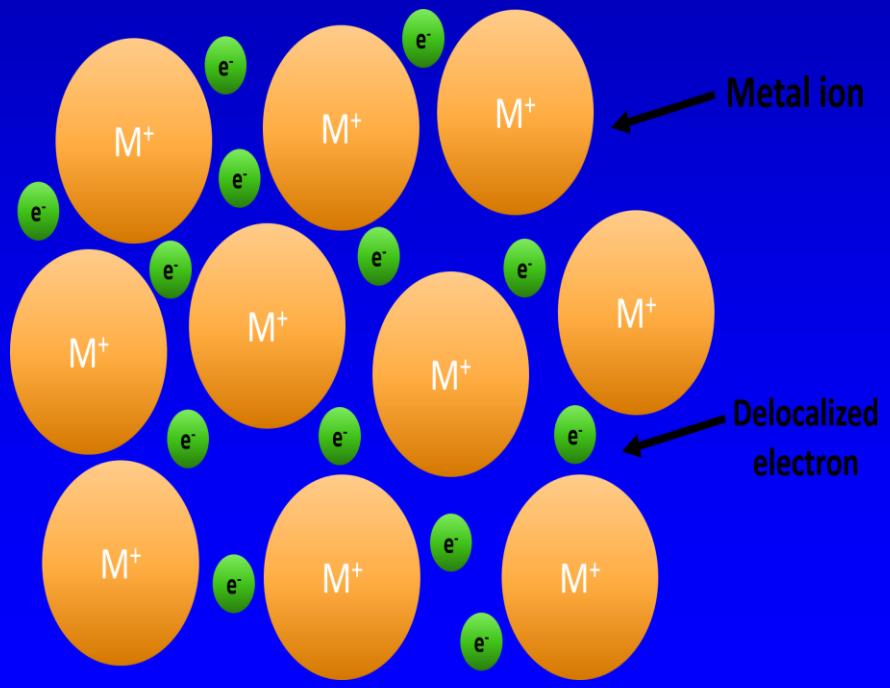
2- Covalent bond رابطة تساهمية

A covalent chemical bond results from the sharing of electrons between two atoms with similar electro negativities.



3-Metallic bond الرابطة المعدنية

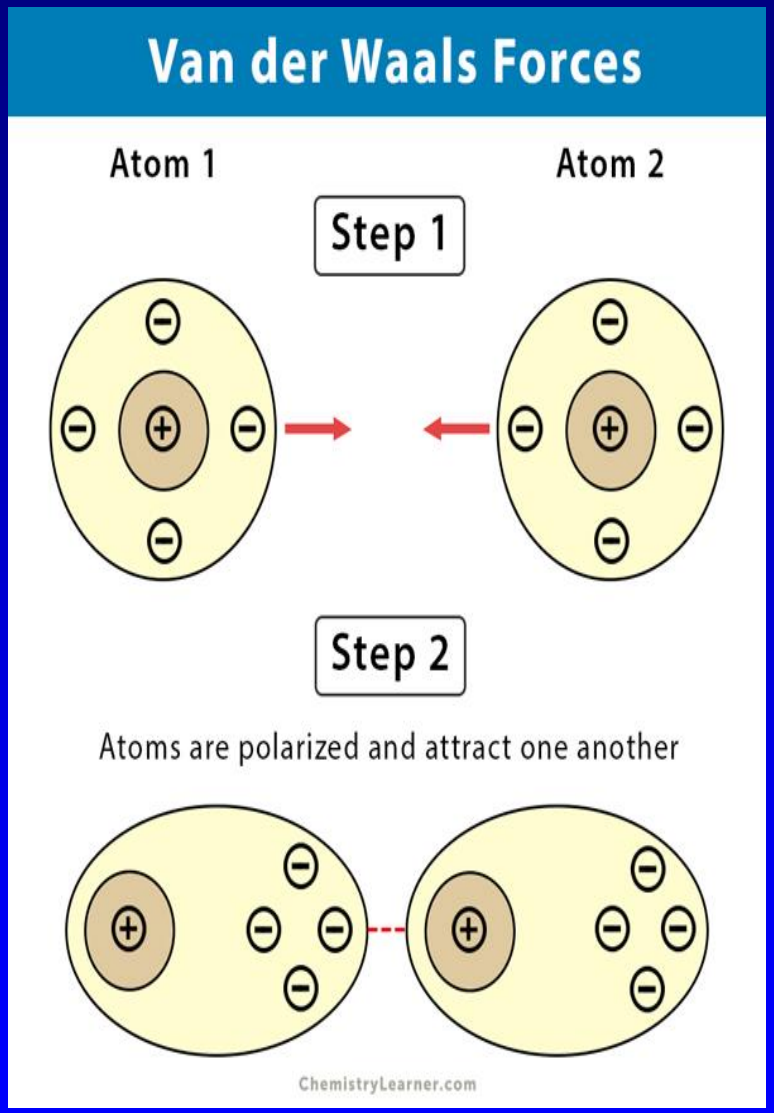
Similar to covalent bond, except innermost electrons are also shared. Electrons act like a flowing "sea", moving about the positively charged cores of the metal atoms. This bond is excellent of conductivity, malleability, and ductility, due to the electrons can move freely.



4- Van der Waals bond

رابطة فان ديرفال

Van der Waals forces are relatively weak compared to covalent bonds. It is the residual attractive or repulsive forces between molecules or atomic groups that do not arise from a covalent bond. Usually results in a zone along which minerals are breaks easily (i.e., cleavage as graphite).

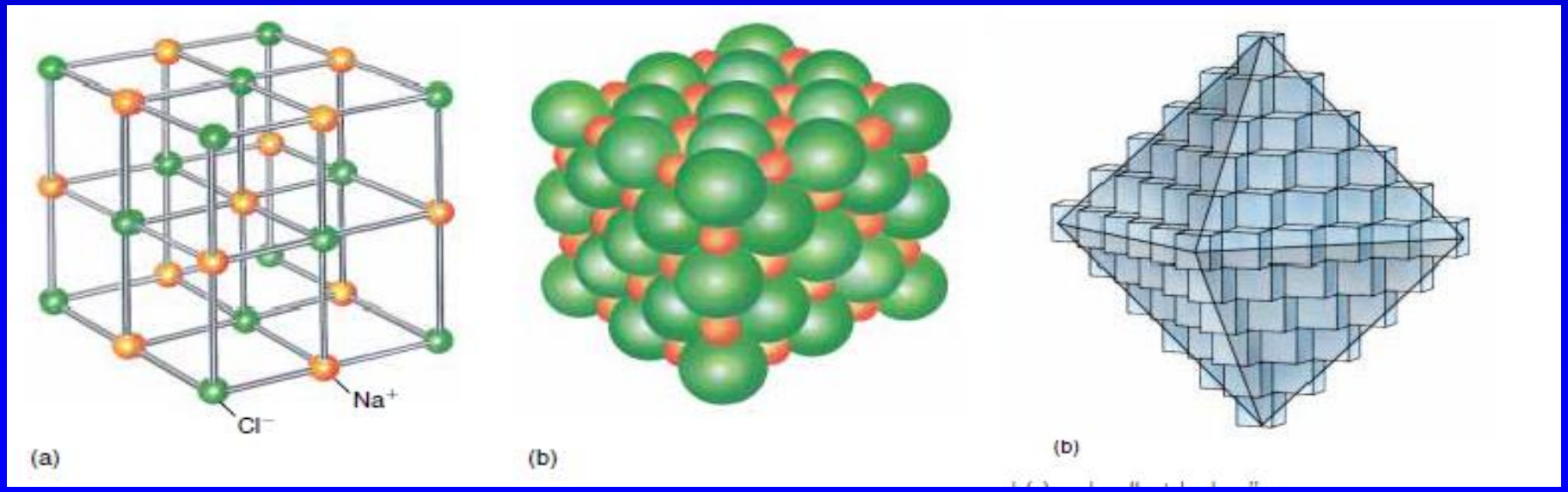


-A total of 88 elements occur naturally in the Earth's crust. However, eight elements make up more than 98 percent of the crust.

Oxygen, O_2
Silicon, Si
Aluminum, Al
Iron, Fe
Calcium, Ca
Magnesium, Mg
Potassium, K
Sodium, Na

THE CRYSTALLINE NATURE OF MINERALS

- A **crystal** is any substance whose atoms are arranged in a regular, periodically repeated pattern. All minerals are crystalline. The mineral halite has the composition Na Cl : one sodium ion (Na) for every chlorine ion (Cl)



Physical properties of minerals

1. Optical properties خواص بصرية

Color

Color is a diagnostic property for identifying minerals.

The color is idiochromatic or allochromatic

وحيدة او متعددة التضوء

يعتمد على طول الموجات الضوئية التى تنعكس من المعدن

Streak
المخدش

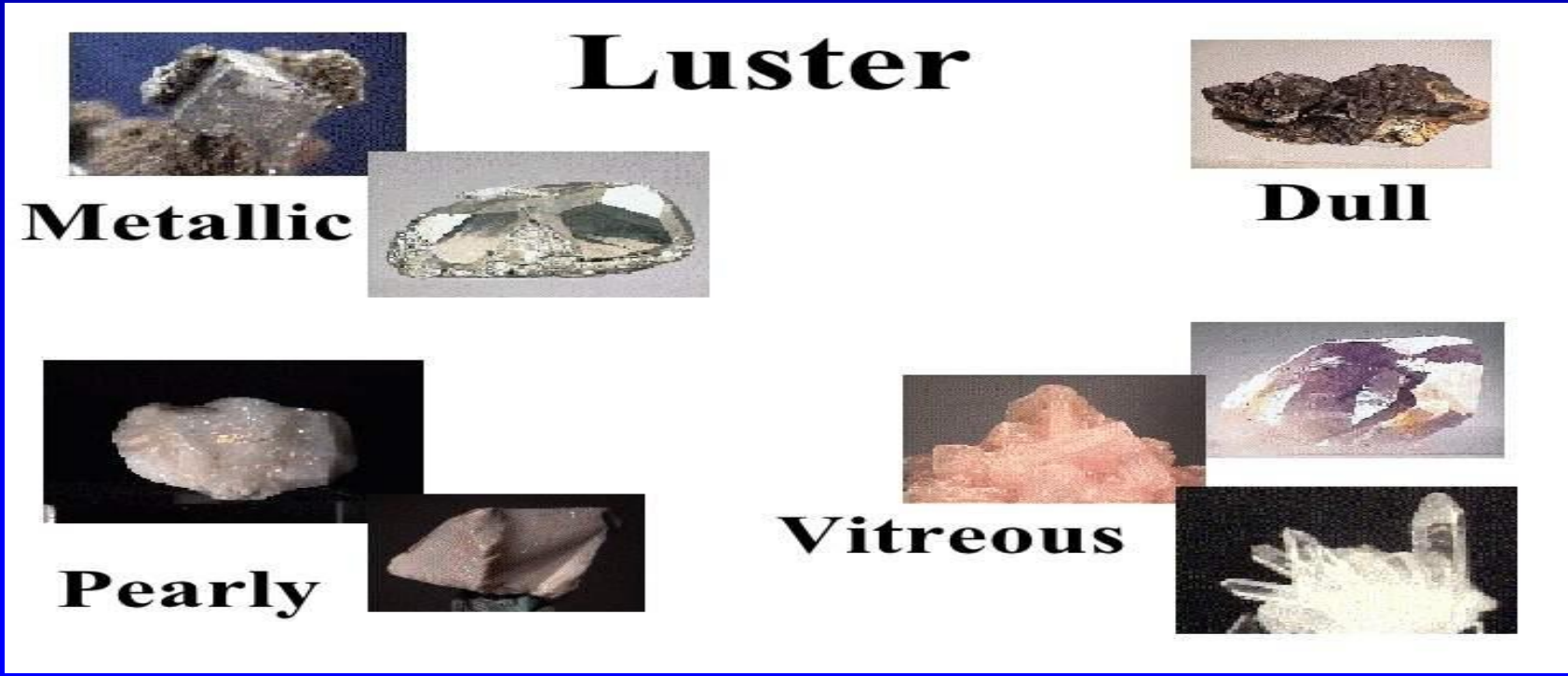
لون مسحوق المعدن
is the color of the powder produced by rubbing the mineral

البريق *Luster*

مقدار ونوع الضوء المنعكس من سطح المعدن

a metallic luster فلزى. Nonmetallic لافلزى

Glassy (quartz), pearly (talc) لؤلؤي, earthy (kaoline) ارضى, and resinous كهرمانى (sulfur)

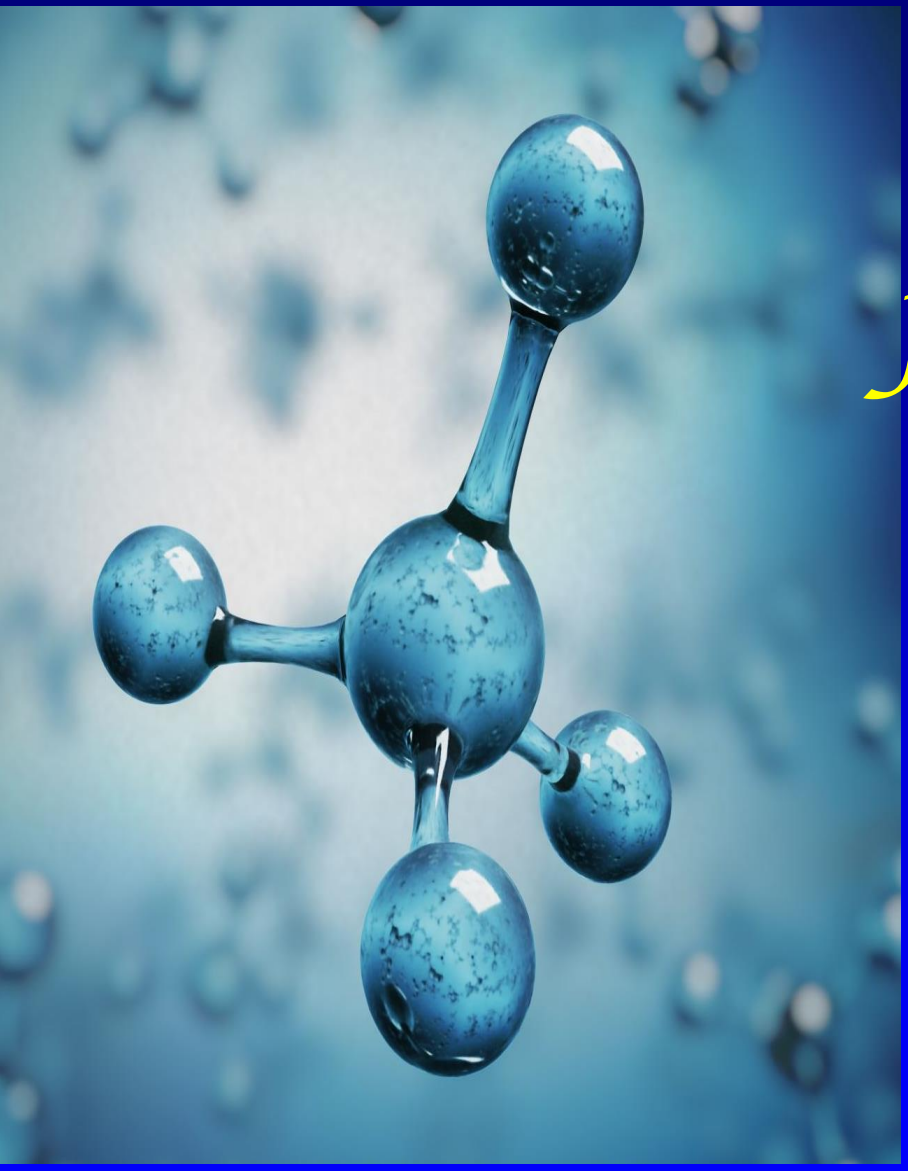


Transparency الشفافية

The amount of light transmitted or allowed to pass through minerals.

قدرة المعدن على انفاذ الضوء

-transparent -opaque



Physical properties of minerals

2. Cohesive properties

خواص تماسكية

1-Tenacity: خاصية الطرق والسحب

the resistance of mineral to breaking, crushing, bending, cutting. It is a mineral cohesiveness.

C- sectile: قابل للكسر the mineral that can be cut by knife into thin pieces (graphite and gypsum).

D- ductile: قابل للطرق a mineral that can be drawn into wires (silver, copper).

E- flexible: قابل للانثناء a mineral that bends but retains it bent form. Does not resume its original shape(clay minerals, asbestos, chlorite).

V- elastic: مرنةmineral that after bending returned to its original position (Muscovite)

Hardness الصلادة

المقاومة التى يبديها المعدن تجاه الخدش

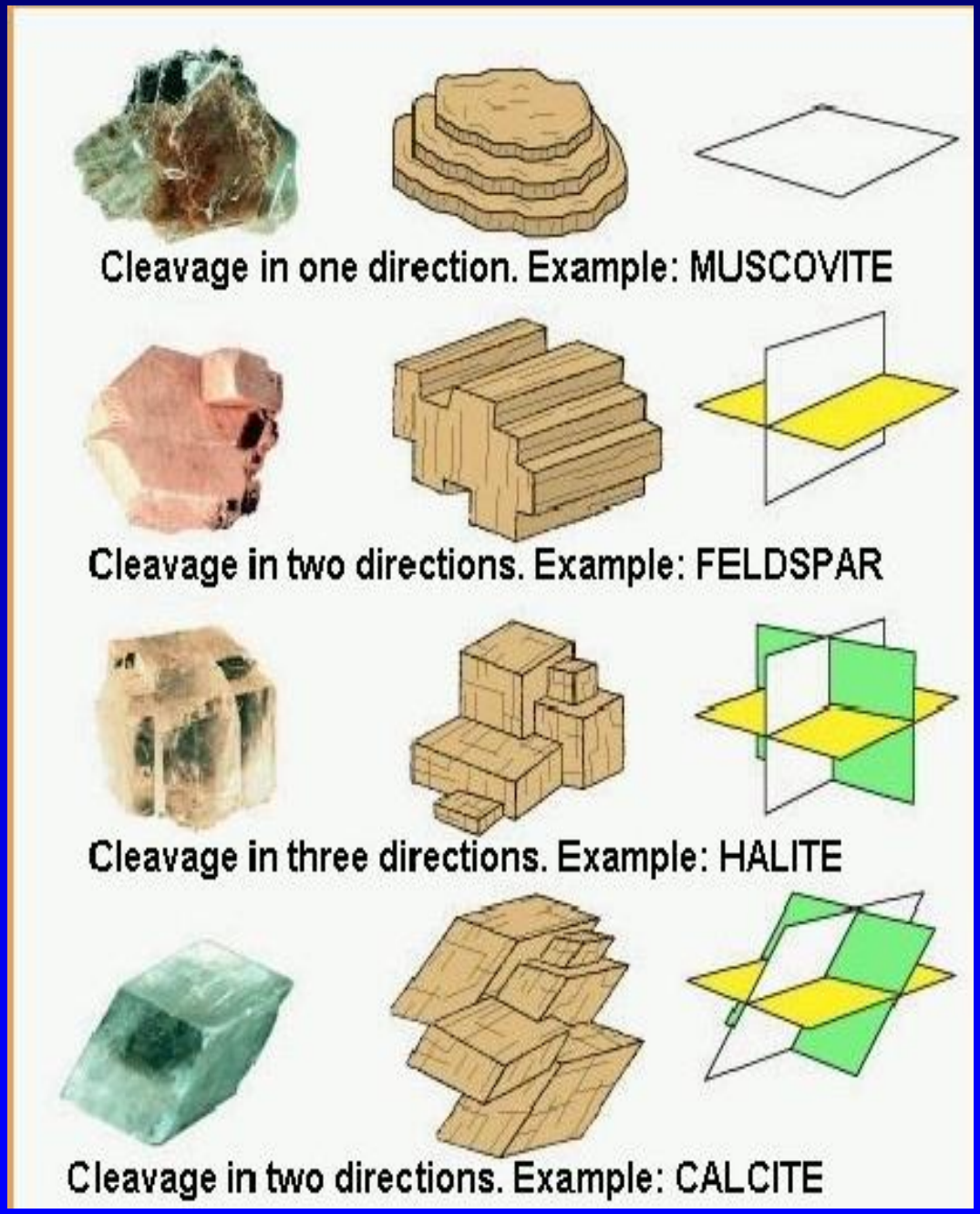
is the degree of mineral resistance to scratching.

Mohs Hardness Scale

Cleavage
الانفصام

is the tendency of a mineral to break along planes of weak bonding (related to crystal structure).

انفصال المعدن بسهولة في اتجاهات معينة نتيجة للتركيب الذري الداخلي للمعدن



Fracture
المكسر

is the shape of minerals that don't have cleavage when broken.

نوع السطح الناتج من كسر المعدن

Types of Fracture

- Conchoidal – curved
- Irregular – uneven
- Fibrous – splintery
- Hackly – jagged-edged

Other Properties

1-Crystal Form الشكل البلورى

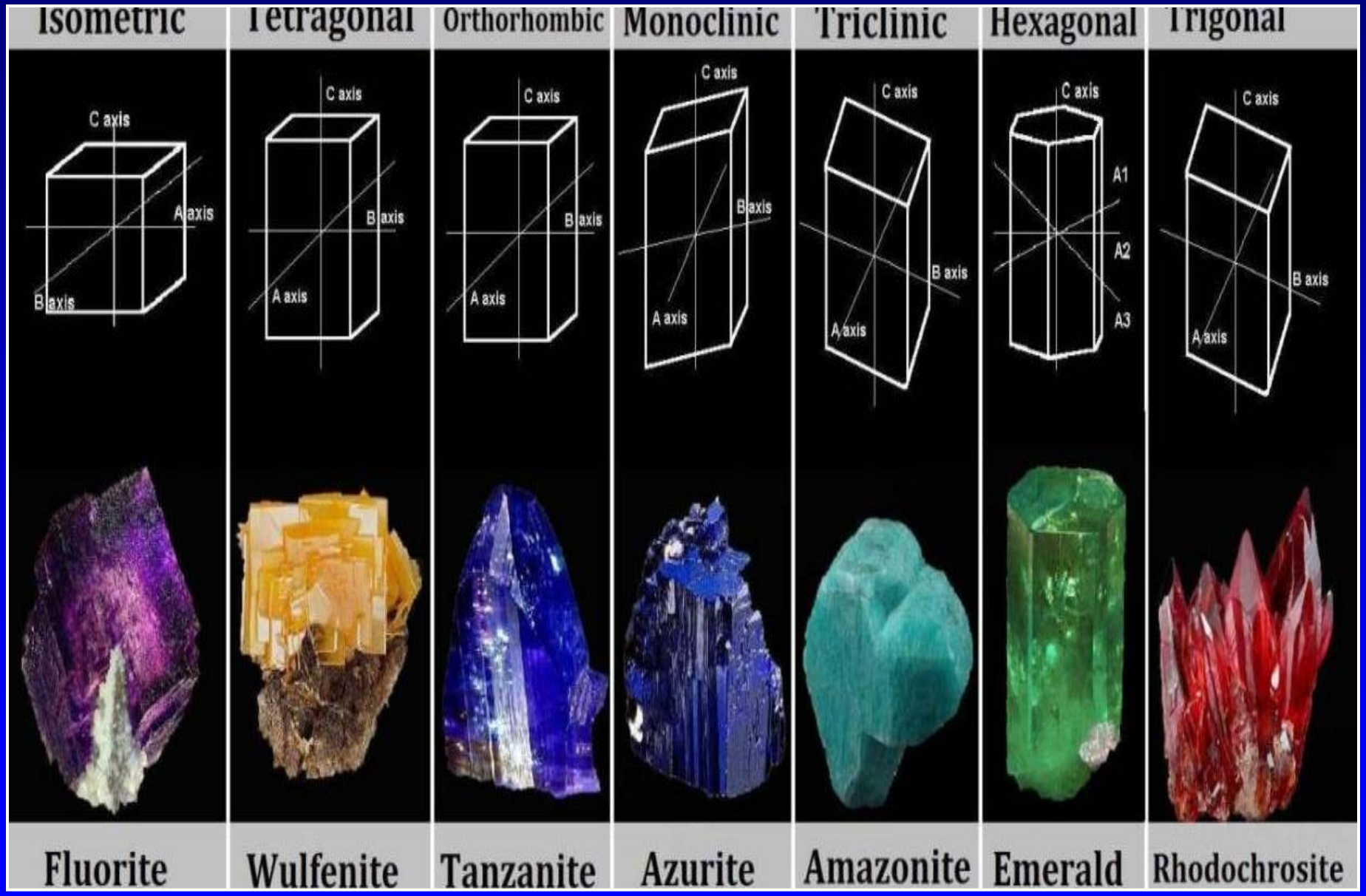
- -is the characteristic shape of a mineral
- -cool of magma and arrangements of atoms
- -Single or aggregates

(a)

Quartz Prismatic منشوری

massive کبیر

Crystal systems



2-Specific gravity الوزن النوعى

كثافة المعدن بالنسبة لكثافة الماء

is a number representing the ratio of the weight of a mineral to the weight of an equal volume of water.

For example if a mineral weighs three times as much as an equal volume of water, its specific gravity is 3.

3-Electrical and Magnetic properties:

خواص حرارية ومغاطيسية

Some minerals possess interesting electrical properties:

Pyroelectricity: الكهربية الحرارية minerals become electrified when subjected to a marked change in temperature, e.g. tourmaline.

Piezoelectricity: كهربية ضغطية pressure induces electrical charge, e.g. quartz.

Magnetism: خواص مغناطيسية

the property of minerals that allows attracting or repelling other magnetic materials, e.g. magnetite, pyrrohtite.

Paramagnetic: attract other magnetic minerals

Diamagnetic : repel other magnetic minerals

6- Taste:الطعم halite has a salty taste.

7- Smell: الرائحة sulfur streak smells like rotten eggs as Pyrite (SO2), Arsenopyrite (Garlic**)** ثوم.

8- Feel: الملمسTalc feels soapy صابونى and graphite feels greasy.

9- Chemical reactions to HCl: carbonates such as calcite effervesce (fizz) with HCl. While, dolomite CaMg (C03)2 must be scratched and powdered to fizz.

10-Play of colour, عرض الالوانby turning mineral against light it appear to have several colours, labradorite, opal, diamond.

11-Fluorescence:التضوء Some minerals glow' in the dark under a black light (U.V. light) due to excitation of electrons.

12-Luminescence: التفلرsome minerals glow or emit light when heated or exposed to in dark to ultra-violet or x-ray, e.g. fluorite (CaF2) , opal

13-**Phosphorescent,التفسفر** luminescence continuous even after excitation.

Mineral Groups

Minerals are classified according to anions into the following:

1-تجميع كل المعادن التى يكون عنصر ما جزء اساسى في تركيبها

معادن الحديد: Fe_3O_4 (magnetite), FeS_2 (pyrite), Fe_2O_3 (hematite), $FeCO_3$ (sedrite), $Fe_2O_3.nH_2O$ (limonite)

٢- تقسم المعادن كيميائيا حسب مجموعات الانيون اي حسب الشق الحامضي

SO_4 (2-), CO_3(-2), Cl (-), O_2 (-2), S (-2)

1-**Native elements** عناصرعنصرية metal): native gold (Au), native copper (Cu), native sulfur (S), native silver (Ag), graphite (C), diamond (C).

2-**Silicates** : quartz (Si02), potassium feldspar (KAlSiO8), Ca- Plagioclase feldspar (CaAbSi20g), Na-plagioclase feldspar (NaAlSiO8).

3-**Carbonates**: calcite ($CaC0_3$), dolomite (CaMg(C03)2).

4-**Oxides**: water ice (H_20), hematite (Fe203), magnetite (Fe_30_4), corundum (AI2O3).

5-**Halides**: halite (NaCI), fluorite (CaF_2).

6-**Hydroxides**: limonite (iron hydroxide), bauxite (aluminum hydroxide).

7-**Sulfates**: gypsum ($CaS0_4$.2H2O), anhydrite ($CaS0_4$), barite ($BaS0_4$).

8-**Phosphates**: apatite.

9-**Sulfides**: pyrite (FeS2), galena (PbS), sphalerite (ZnS).

Other - borates, uranates.